

# Alexandru Nicolae Serban

Mechatronics  
Engineering Graduate



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📍 Alba Iulia, Romania

## PROFILE

Mechatronics Engineering graduate with hands-on experience in **robotics, control, and computer vision**. Skilled in **C++, Python, ROS2, and embedded systems**, with internships in drone control and modular robotic platforms. Strong track record of delivering **real-world robotics solutions from concept to deployment**.

## SKILLS

**Programming and Tools:** C++, Python, ROS2, Matlab/Simulink, Embedded (Jetson, Raspberry Pi, Arduino), Git, Linux, Ladder Logic, Structured Text, Function Block Diagrams

**Robotics and Control:** MPC, PID, EKF, SLAM, Path Planning, Inverse Kinematics

**Hardware and CAD:** SolidWorks, Creo, Onshape, Sensors, Actuators, 3D Printing

**Machine Learning and Vision:** Tensorflow, Yolo, Autoencoders, OpenCV, SORT

**Soft Skills:** Teamwork, Fast-Learner, Problem-Solving, Flexible, Creative Thinking, Team Player

## CERTIFICATES

- SolidWorks, CSWP Certificate, Fontys University of Applied Sciences

## EXPERIENCE

### Full-Time Software and Robotics Engineer Intern

Affix Engineering

02/2025 – 07/2025 | Best, Netherlands

Project: Universal software for flex feeder systems

- Built a **Python vision system** with YOLO-NAS-S + Autoencoder for real-time defect detection (**mAP ≥ 0.8, <2s inference**).
- Designed a **GUI for dataset creation and model training**, reducing setup time by **70%** for non-technical users.
- Implemented **modular training/augmentation classes**, cutting code maintenance time by **30%**.
- Engineered a **magnetic ISO 909 tool changer** for SCARA robots, enabling tool swaps in **<5 seconds** with **100% repeatability**.

### Full-Time Software and Control Engineer Intern

MultiRotorResearch

09/2023 – 02/2024 | Eindhoven, Netherlands

**Project:** Research and implementation of trajectory control for large multi rotor vehicles

- Developed a **C++ MPC library** on Nvidia Jetson, reducing trajectory tracking error by **25%** compared to PID-only control.
- Integrated controller with Python, enabling **real-time drone communication with <50 ms latency**.
- Tuned low-level PID loops, improving stability under wind disturbances in **simulation and real flights**.
- Applied multiprocessing, boosting controller responsiveness and ensuring **smooth flight in long trajectories**.

### Technical Peer Mentor

FTC Team Xeo 14278

01/2024 – Present | Alba Iulia, Romania

- Mentored **10+ students** in CAD, programming, and design, raising competition readiness and team performance.
- Provided design reviews and hands-on guidance, reducing build iteration cycles by **40%**.

### Mechatronics Engineer

FRC Team Pi 6968 and FTC Team Xeo 14278

06/2022 – 07/2023 | Eindhoven, Netherlands/ Alba Iulia, Romania

- Designed and prototyped **swerve drive & modular intake systems**, improving maneuverability and scoring rate.
- Led mechanical design workflow and 3D printing pipeline, cutting prototyping time from **12 weeks** to **3-4 weeks**.
- Programmed robot subsystems in **C++ OOP architecture**, reducing debugging time by **50%**.

## EDUCATION

### Bachelor's Degree in Mechatronics Engineering

Fontys University of Applied Sciences, Eindhoven, Netherlands

09/2021 – 07/2025 | Eindhoven, Netherlands

- Minor in **Adaptive Robotics - Principles of Robotics, Vision, Mapping and Localization, ROS2, Hardware Abstraction and AI**
- **Projects:** Smart Factory **Digital Twin** in Isaac Sim; **ROS2 differential robot** for mapping and localization; Sensor Fusion **ROS2** node; Factory automation on **Siemens PLC**; Object detection and image segmentation for **RoboCup** competition